Day - 3

04 November 2026 09:40

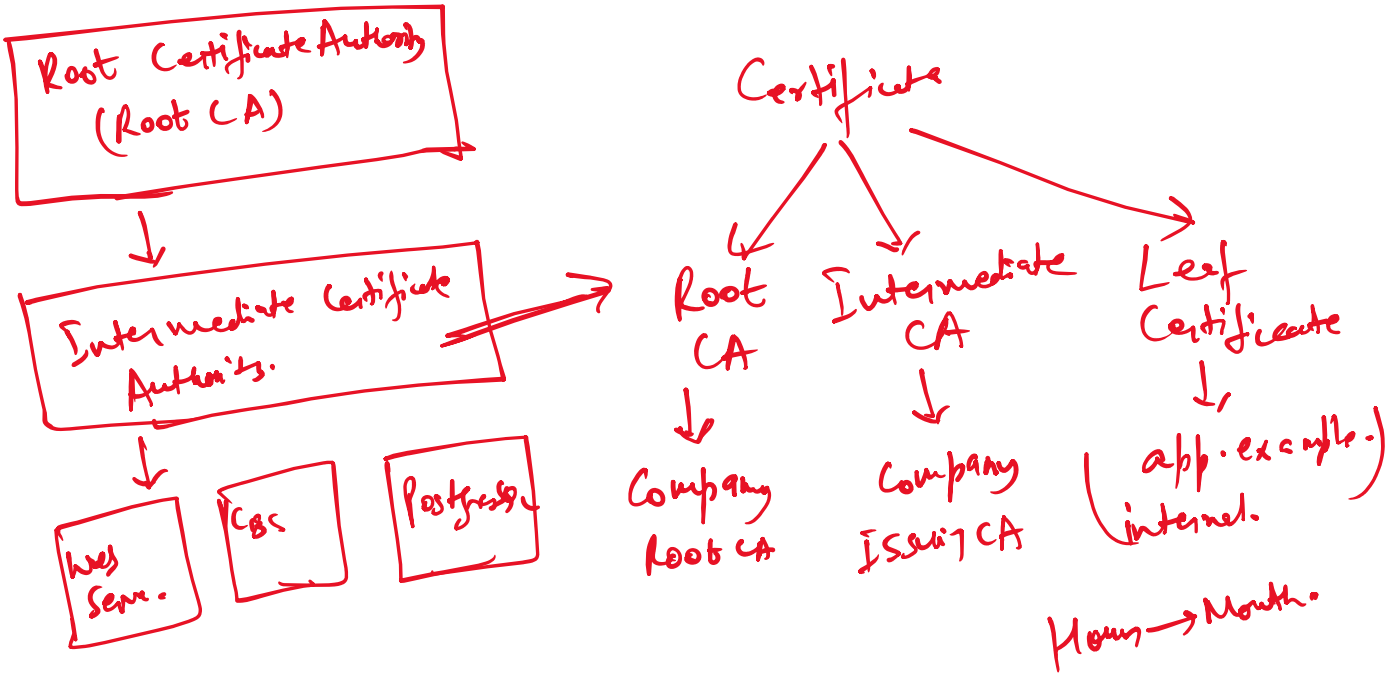
1. ~~PKI Secret Engine~~ -
2. ~~Transit Secret Engine~~.
3. Auto-Unseal & Seal Wrapping.
4. Vault Agent & Vault Agent templating.
5. HA, Replication & DR.
6. Monitoring, Audit Devices & Best Practices.

1. PKI Secret Engine :- Automate the generation, signing & Mng. of Digital certificate.

↓

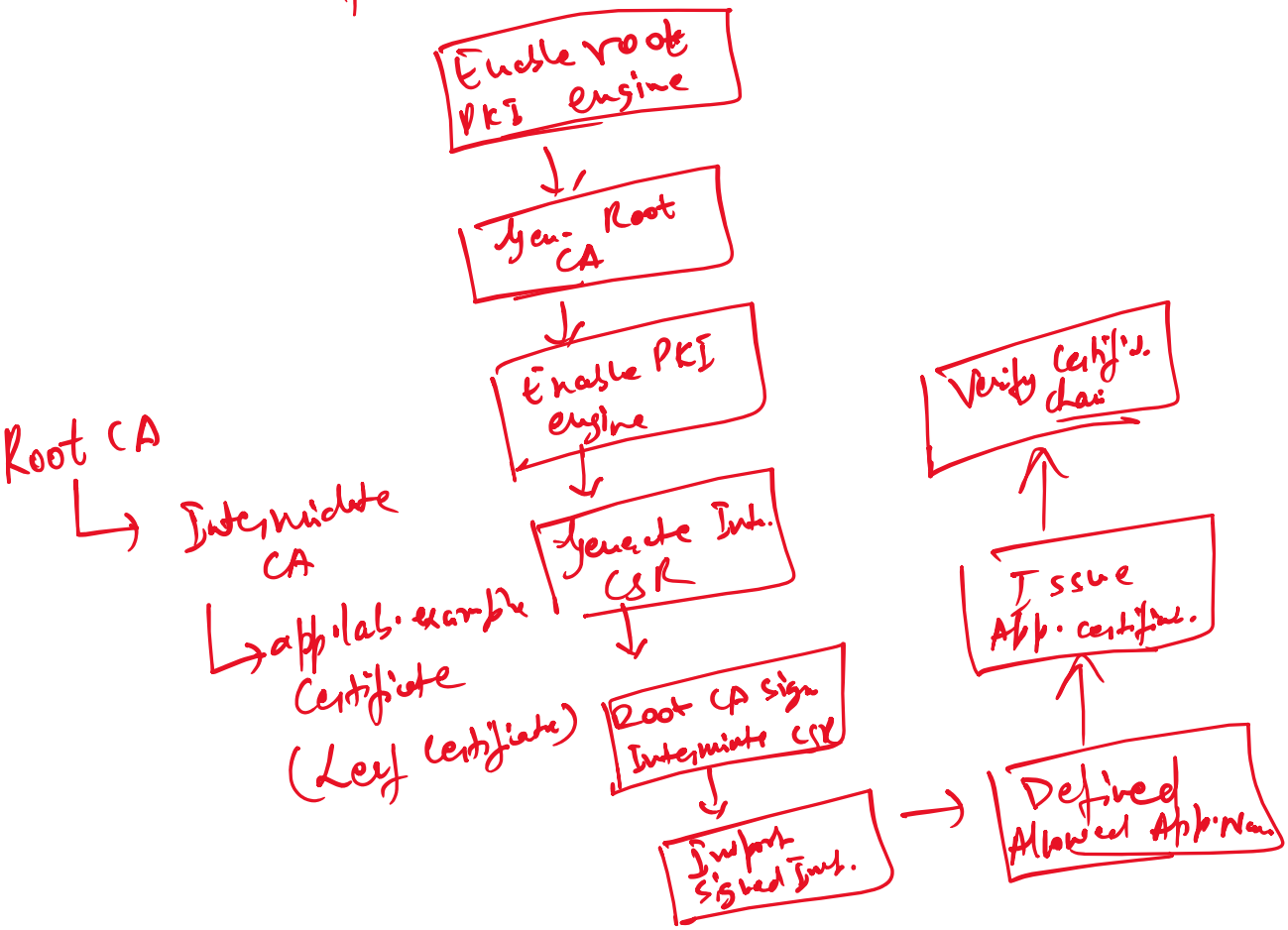
Public Key Infrastructure

1. Private Key gen.
2. Create CSR
3. Req. Certificate.
4. Signing
5. Distributed Certificates.
6. Replace them B/f expiry.



1. Common Name / Subject (CN)
2. SAN (Subject Alternative Name)
3. Issuer
4. Serial No.
5. Validity
6. Public Key.
7. Digital Signature.

Build a private Certificate hierarchy & Issue a certificate for an Application.



vault write pki-int/roles/training-web allowed_domains="lab.example" allowed_subdomains=true allow_bare_domains=false

vault write pki-int/roles/training-web allowed_domains="lab.example" allowed_subdomains=true allow_bare_domains=false allow_ip_sans=false server_flag=true client_flag=false max_ttl="24h"

"domain"

Server authentication.

Client Authentication.

→ Limit certificate lifetime.

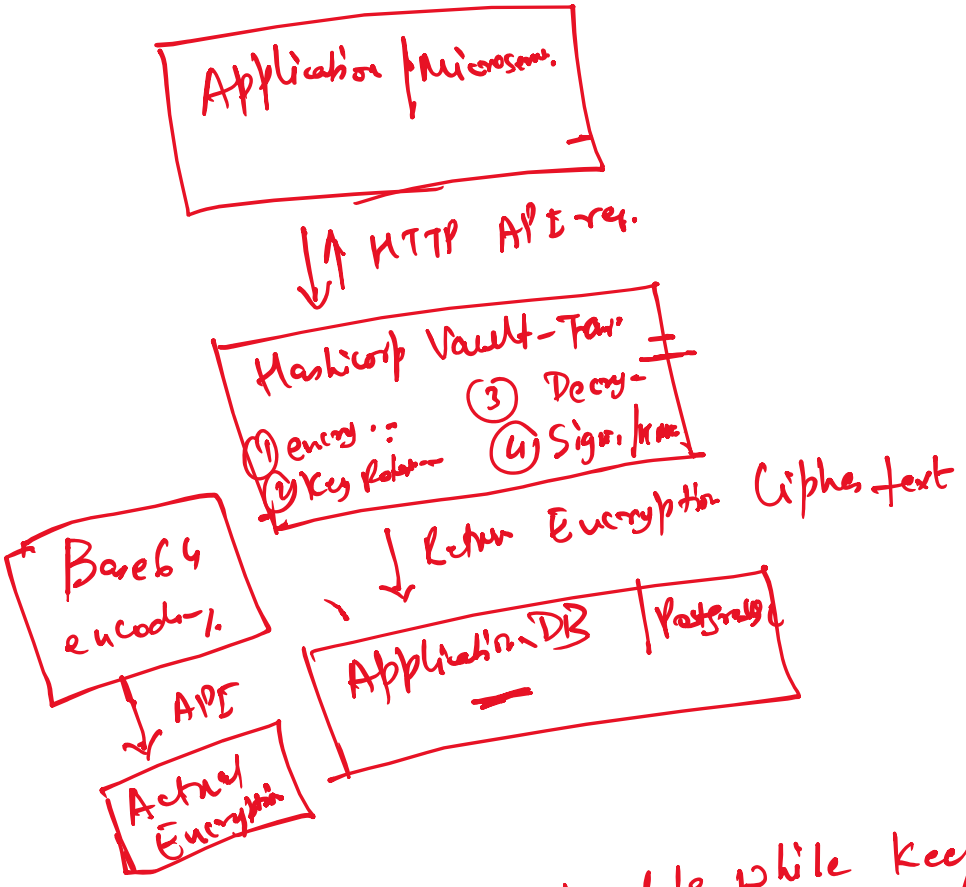
lab.example.

Valid root → Intermediate → Application ~~Depth~~ Chain.

(2.) Transit Secret Engine – Encryption As a Service:–

encry.
Decry.
Sign.
Verify.

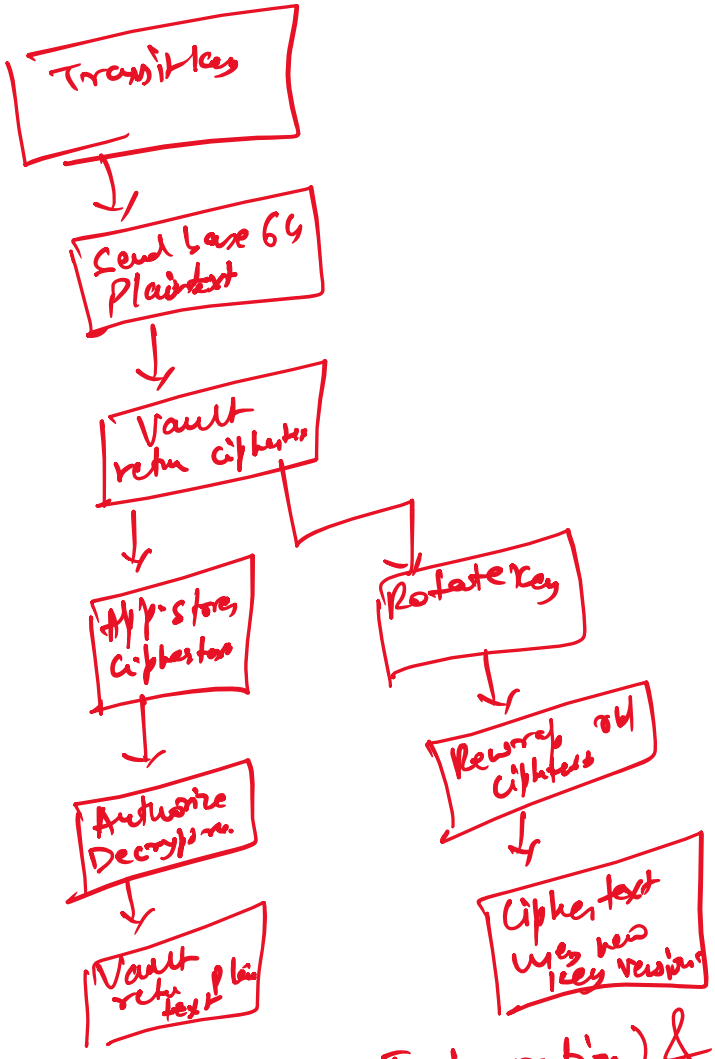
→ Encryption As a Service (EaaS)



to encrypt App. data while keeping

Use Vault to encrypt App. data ...

the encryption key inside Vault.



(3.) Auto-Unseal (cloud KMS Integration) & Seal Wrapping :-

AWS, Azure, Azure Key Vault or an HSM

Auto Unseal –

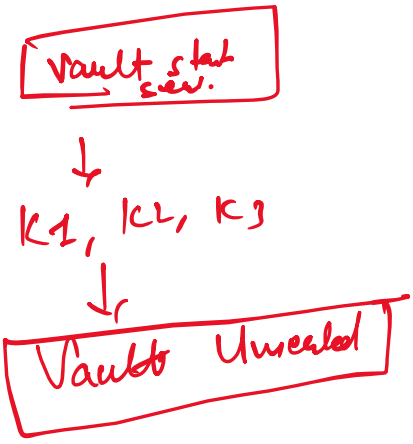
→ Automatically unlocks Vault's encryption barrier when server starts.

Seal Wrap – Add another layer of encryption to selected sensitive data stored inside vault.

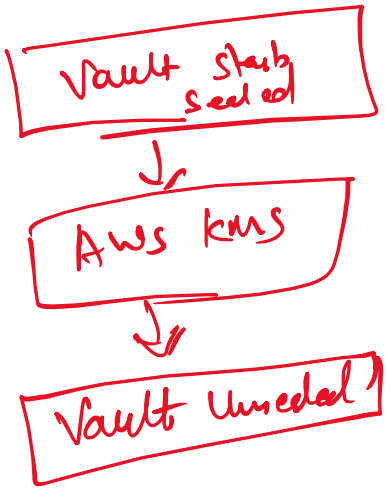
... have shamir Unseal shares.

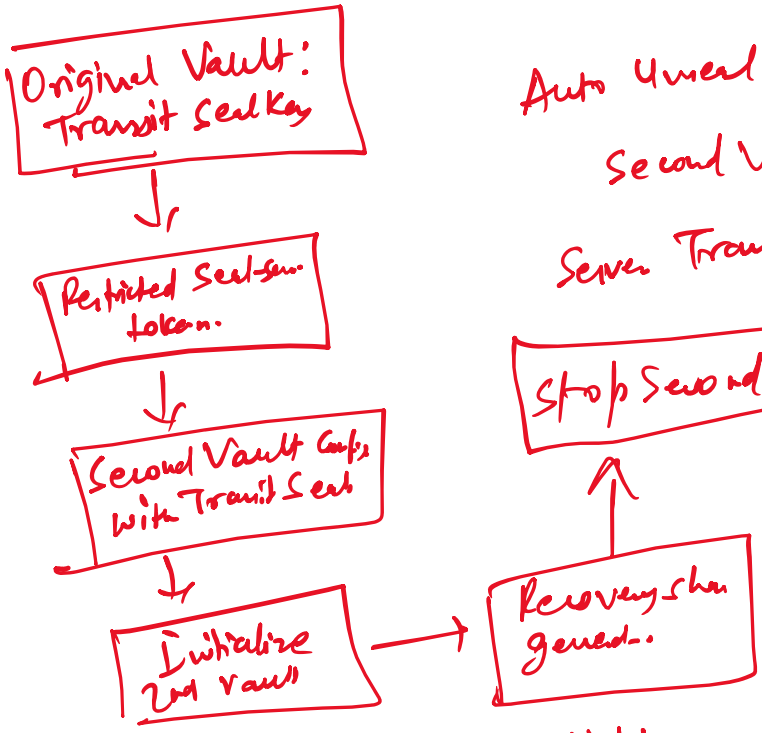
Restart → 3 unique shamir Unseal shares.
↳ Manual Unseal.

Manual Unseal



AWS → Auto Unseal



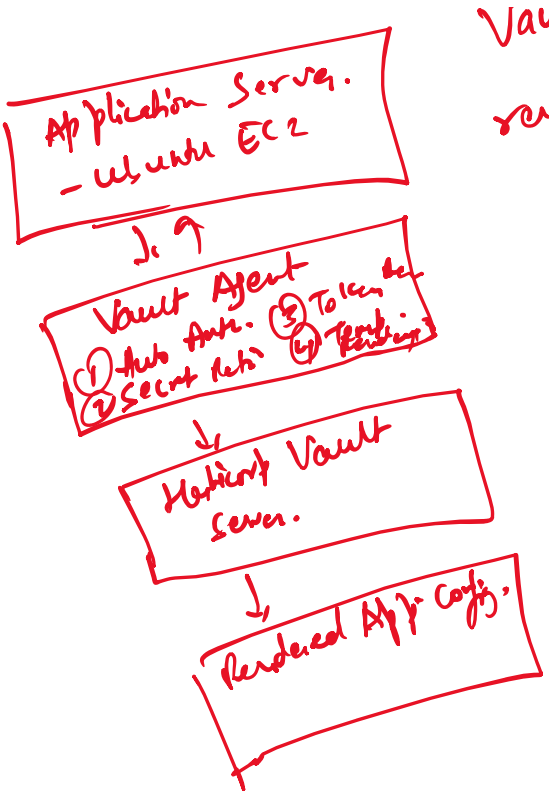


Auto Unseal locally with a second Vault Serv. & first Server Transit engine.

* Vault Agent & Vault Agent Template:-

Client-side process (Demon) → Appl. authenticate to HashicorpVault, retrieve secrets, manage Authe.

Vault Agent Authenticate AppRole & a Config. file for an Appl.



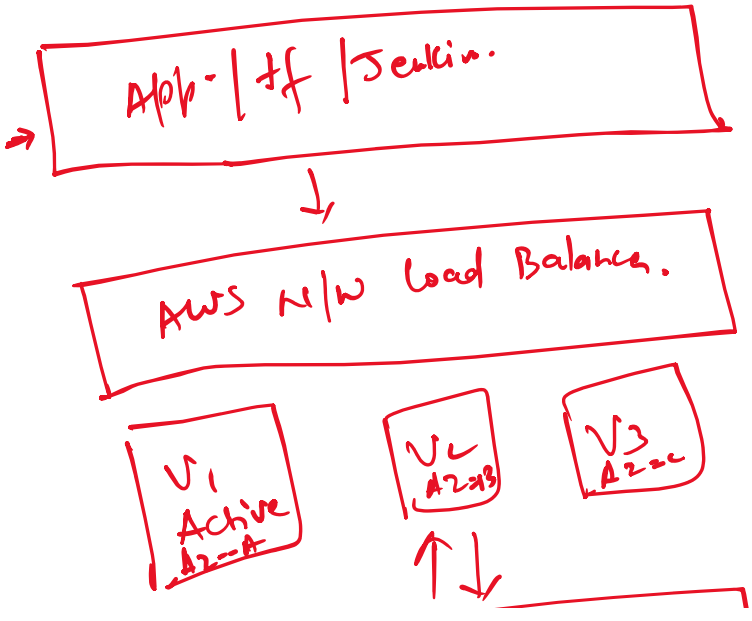
Vault ... render a Config. file for ...

High Availability, Replication & Disaster Recovery:—

(1) High Availability → Vault server crashes? → Min. Downtime.

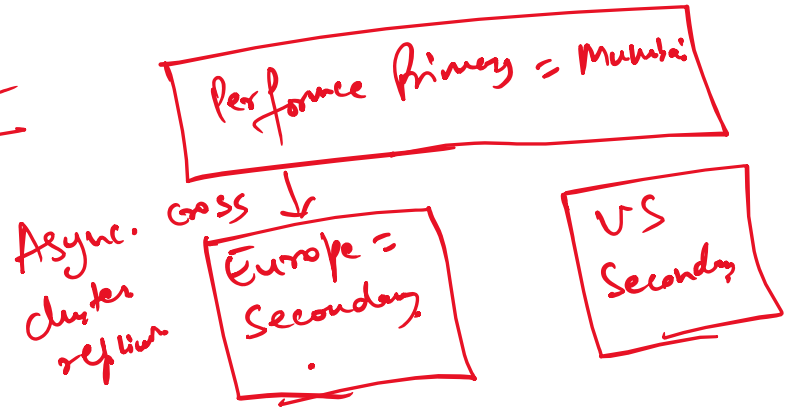
(2) Replication. → Vault data across node, region → scale req. or clusters.

(3) Disaster Recovery → entire vault cluster or DataCentre become unavailable! → Major failure.

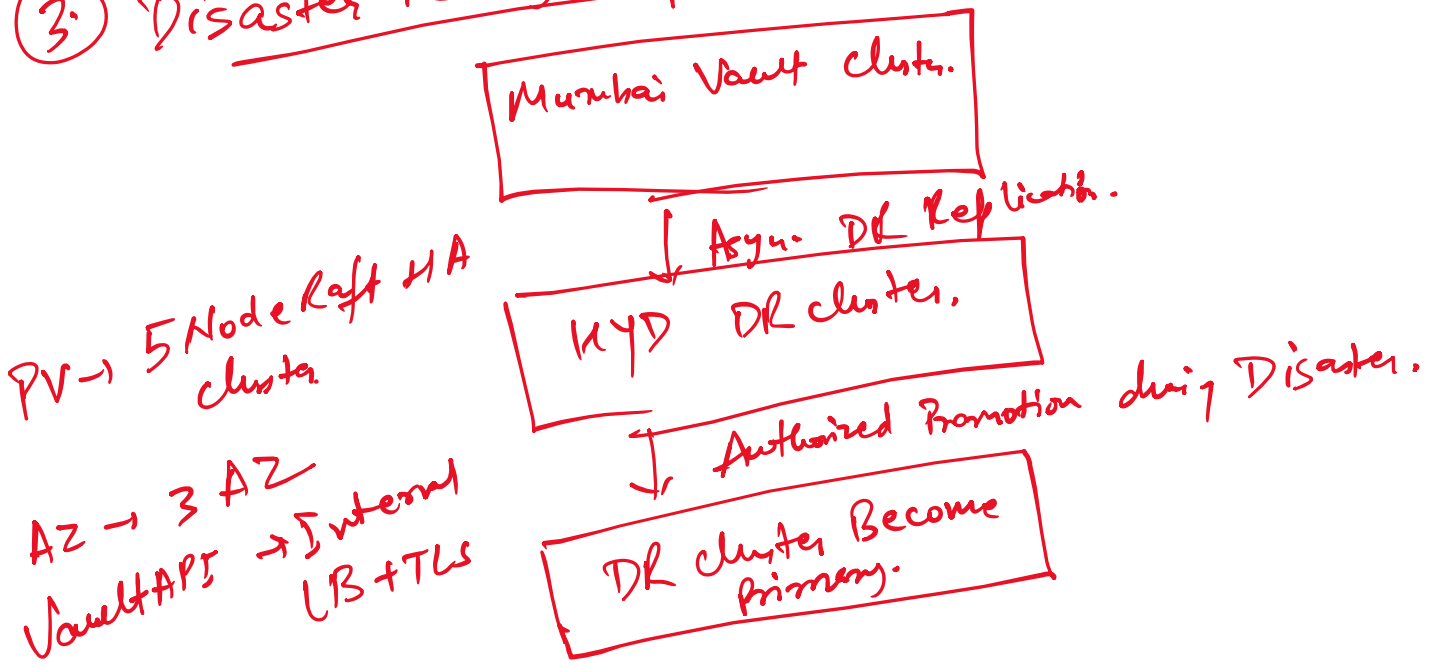


Raft integrated storage.

(2.) Replication :-



(3.) Disaster Recovery Replication :-



Monitoring → CPU, Memory, Req. latency,

Operational Logs → Startup, seal, storage.

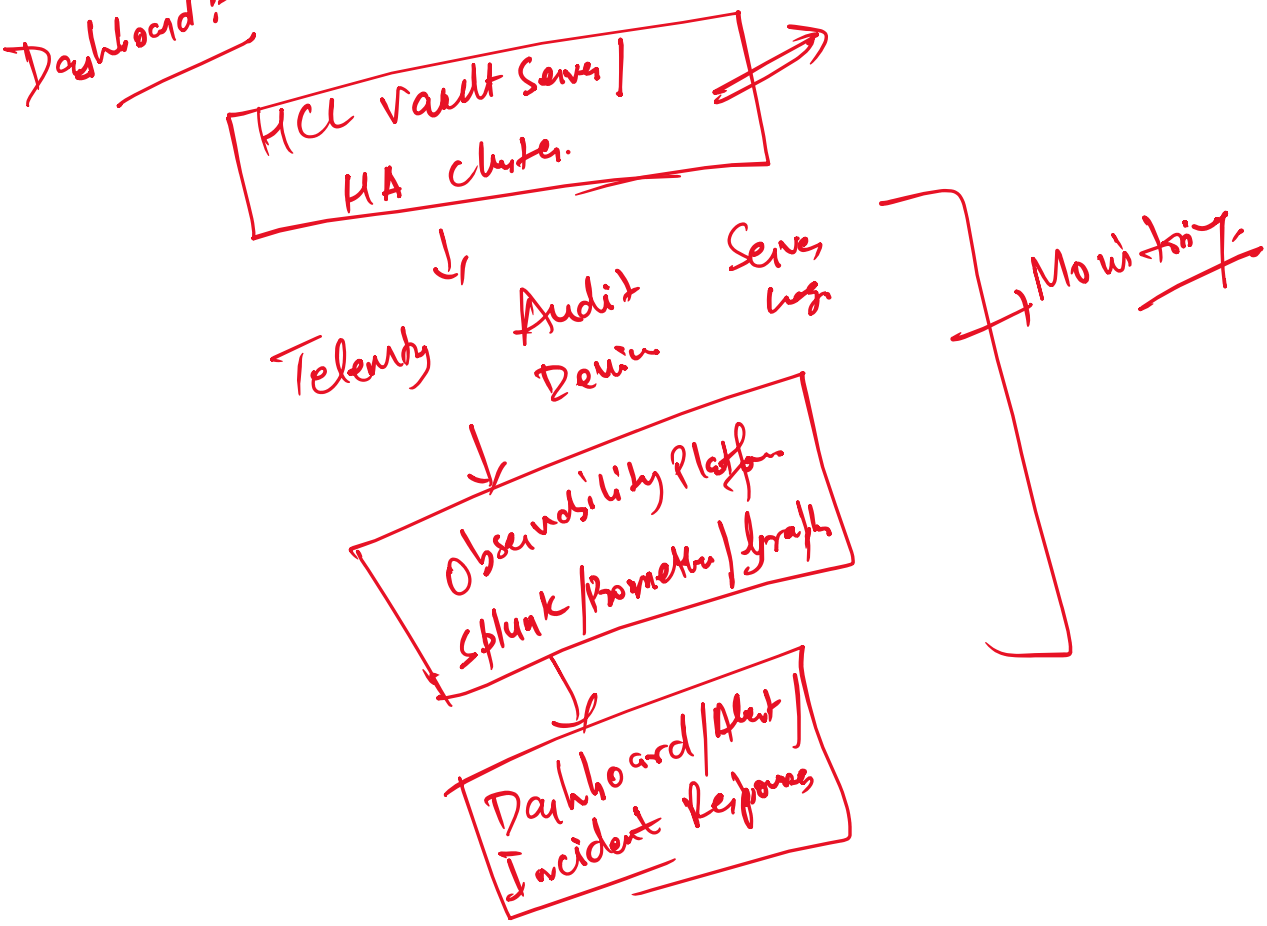
Audit Logs → who? when? & where?

Rejected Auth. failure, Vault sealed.

warning, error, Raft status.

Alert:→ Repeated Auth. f... error, ...

Dashboard:→ Vault Availability,



App. API req.

↓

Vault Authenticate & Authorize

↓

Req. Audit Processing → File | Syslog | HMAC | Socket.

↓

SIEM / Monitoring Tool